

For questions 1-3, identify whether the statement is true or false. If it is false, rewrite the statement to make it true.

1. Line plots are useful to show large quantities of data. *True or False*
Line plots are useful to show small quantities of data.
2. Histograms and stem-and-leaf plots can show large quantities of data and the overall shape. *True or False*
Histograms and box plots can show large quantities of data and the overall shape.
3. Box plots show the overall shape and information to calculate the 5-number summary. *True or False*
4. A fuel efficiency, in miles per gallon, of 250 cars were surveyed. Which graphical display would best display the data?

- ☐ Box plot
 - ☐ Line plot
 - ☒ Histogram
 - ☐ Stem-and-leaf plot
5. Explain your reasoning for question 4.
250 is a large quantity of data that could easily be grouped into ranges, so it is best to use a histogram.
6. Greg was in the market for a new SUV. He recorded the fuel efficiency for his top 12 choices.

{22.5, 17.2, 18, 20.2, 13.6, 19.4, 14, 20.4, 11.6, 16, 15.9, 20}

Explain which display would be better for the data, a line plot or a histogram.

Since there are no repeating values, it is best for Greg to use a histogram so that he can group his data into ranges.

A survey was conducted to middle school students in order to determine their favorite type of music. The frequency and relative frequency tables below represent the data collected.

Frequency Table

Type of Music	Frequency
Rap	65
Hip hop	87
Rock	32
Other	16

Relative Frequency Table

Type of Music	Frequency
Rap	32.5%
Hip hop	43.5%
Rock	16%
Other	8%

7. In order to create a circle graph, the data from the table is needed.

☐ frequency
☒ relative frequency

8. In order to create a bar graph, the data from the table is needed.

☒ frequency
☐ relative frequency

9. Which of the follow are types of displays for univariate numerical data? *Select all that apply.*

- ☐ Frequency Table
☒ Box plot
☒ Line Plot
☒ Histogram
☐ Circle Graph
☒ Stem-and-leaf plot
☐ Bar Graph

10. Which of the follow are types of displays for univariate categorical data? *Select all that apply.*

- ☒ Frequency Table
☐ Box plot
☒ Line Plot
☐ Histogram
☒ Circle Graph
☐ Stem-and-leaf plot
☒ Bar Graph

Algebra 1
Unit 12
Lesson 2 Homework

Name Answer Key
Date _____
Period _____

Identify whether each type of data is categorical or numerical. Then identify if it is univariate or bivariate.

1. The number of houses sold in Broward County in different price ranges.

☐ Categorical
☒ Numerical

☒ Univariate
☐ Bivariate

2. The types of houses sold in Broward County.

☒ Categorical
☐ Numerical

☒ Univariate
☐ Bivariate

3. The number of houses sold in Broward County different price ranges and the average time the houses are on the market for.

☐ Categorical
☒ Numerical

☐ Univariate
☒ Bivariate

Use the following for questions 4-6.

Greg was in the market for a new SUV. He recorded the fuel efficiency for his top 12 choices.

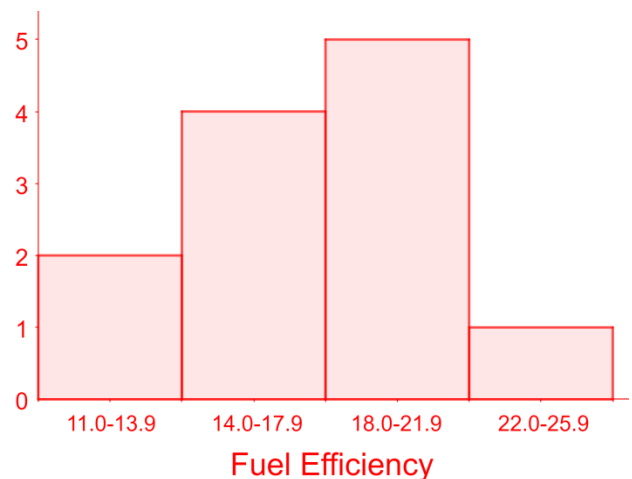
{22.5, 17.2, 18, 20.2, 13.6, 19.4, 14, 20.4, 11.6, 16, 15.9, 20}

4. This data set is ☐ Categorical and ☒ Univariate
☒ Numerical ☐ Bivariate

5. Which is the best graphical representation to display the data?

☐ Scatterplot
☐ Box plot
☐ Line Plot
☒ Histogram
☐ Circle Graph
☐ Stem-and-leaf plot
☐ Bar Graph

6. Create a graphical representation for the data chosen in question 5.



The National Weather Center collected data on weather conditions in various locations. Data was collected over a few years comparing the elevation above sea level in feet and the average number of clear days in 12 different cities in the United States.

City	Elevation above sea level	Average number of clear days
Anchorage, AK	102	45
Boston, MA	141	102
Tampa, FL	48	146
Salt Lake City, UT	4,226	95
Charlotte, NC	761	127
Seattle, WA	174	78
Boise, ID	2,730	115
Milwaukee, WI	617	90
New York, NY	33	72
Cincinnati, OH	482	111
Boulder, CO	5,430	65
San Francisco, CA	52	198

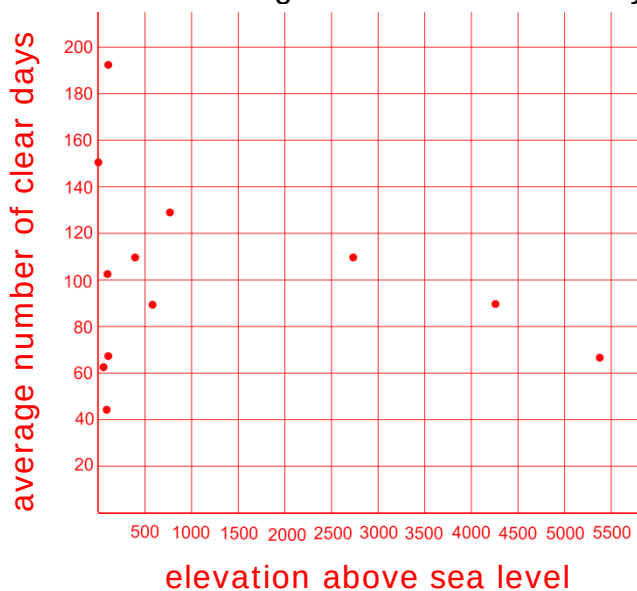
7. The data shown in the table is ☐ univariate.
☒ bivariate.

8. The data in the table should be displayed in a ☐ line plot
☒ scatter plot.

9. Explain your reasoning from question 8.

Bivariate data should be shown with a scatter plot to show the relationship between the two variables.

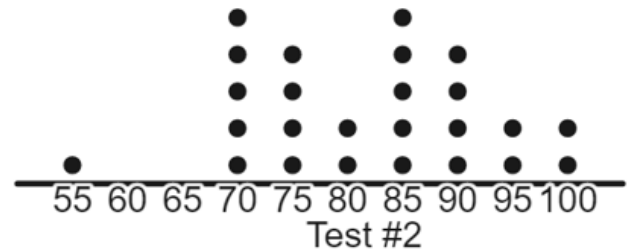
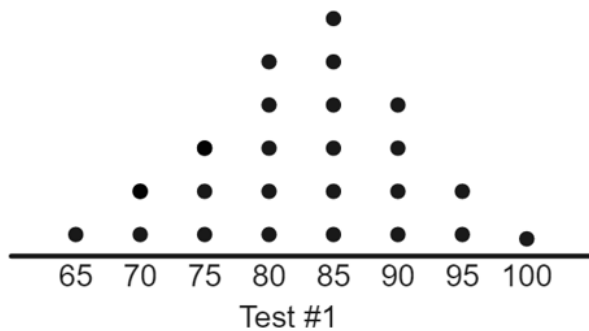
10. Create a display to represent the relationship between elevation above sea level and the average number of clear days.



Algebra 1
Unit 12
Lesson 3 Homework

Name Answer Key
Date _____
Period _____

Two dot plots are shown, with results from two recent tests.



1. What is the mean of the scores for Test #1?

$$\frac{1990}{24} \approx 83$$

2. What is the median for the scores of Test #1?

85

3. Which measure of center is the most appropriate to use to describe the scores of Test #1? Explain how you know.

Since the data has a fairly normal distribution, it is most appropriate to use the mean.

4. Find the interquartile range of Test #1 and interpret it in the context of the problem.

$$90 - 77.5 = 12.5$$

The middle 50% of test scores have a range of 12.5.

5. What is the mean of the scores for Test #2?

$$\frac{2040}{25} = 81.6$$

6. What is the median of the scores for Test #2?

85

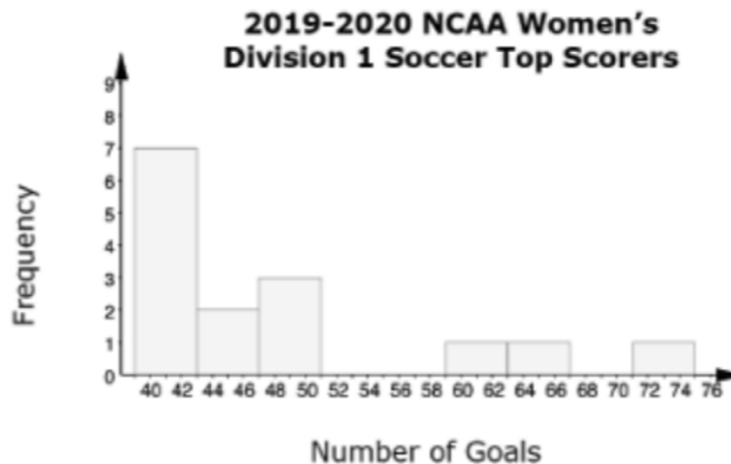
7. Which measure of center is most appropriate to use to describe the scores of Test #2?

Median

8. Write a sentence comparing the results of Test #1 to Test #2 using measures of center and measures of spread.

The results of Test #1 have a fairly normal distribution, so it is best to use mean and standard deviation; the results of Test #2 are skewed, making median and IQR the best choice.

9. In the histogram shown, which measure of center would best be used to describe the data? How do you know?



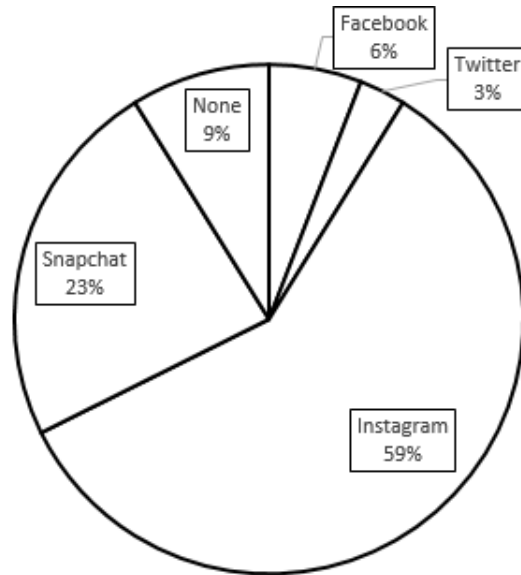
Median – because the data is skewed

10. A table of values for the heights of the students in a class has a mean of 61 inches and a standard deviation of 1.4 inches. Interpret the variation in height.

A standard deviation of 1.4 is fairly low, meaning the majority of students are close to the average height of 61 in.

The students' heights range from 59.6–62.4 inches.

A circle graph is shown, with the results of a survey of a classroom of 34 9th grade students about which social media platforms they use.



1. What is the most frequently used social media platform?

Instagram

2. What is the least frequently used social media platform?

Twitter

3. How many students in the class used Snapchat?

$23\% \cdot 34 \approx 8$ students

4. How many students in the class do not use any social media?

$9\% \cdot 34 \approx 3$ students

5. How many more students use Instagram than Facebook?

$0.59 \cdot 34 \approx 20$

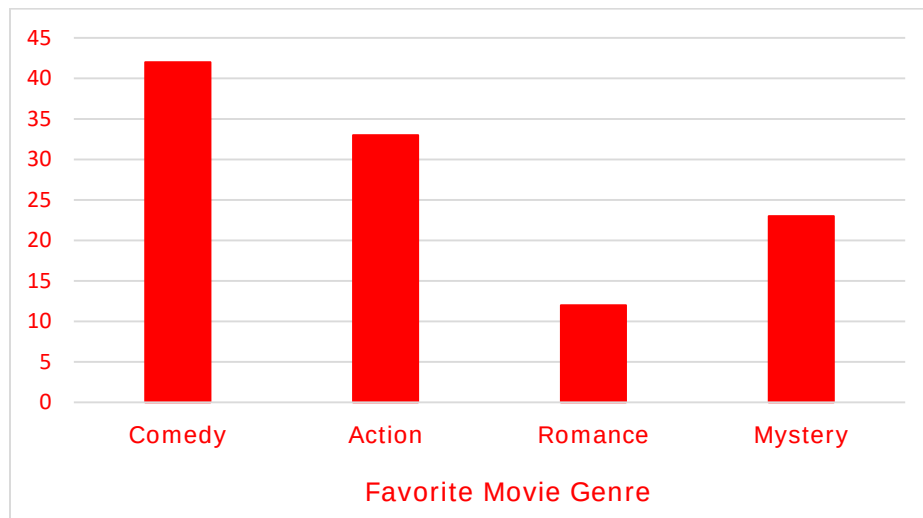
$20 - 2 = 18$ students

$0.06 \cdot 34 \approx 2$

A table is shown with the results of a survey of a group of 110 students about their favorite type of movie.

Favorite Movie Genre	Number of Students
Comedy	42
Action	33
Romance	12
Mystery	23

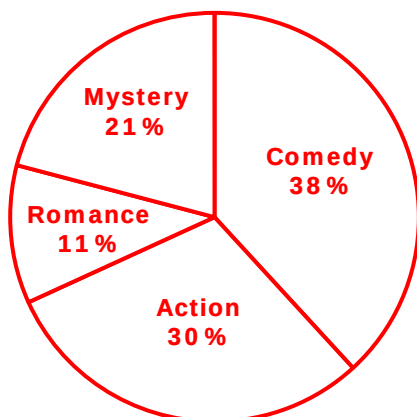
6. Create a bar graph showing the results of this survey.



7. Create a relative frequency table showing the results of this survey.

Favorite Movie Genre	Number of Students	Relative Frequency
Comedy	42	0.38
Action	33	0.30
Romance	12	0.11
Mystery	23	0.21
Total	110	1.00

8. Create a circle graph showing the results of this survey.



9. What percentage of students preferred either mystery or action movies?

51%

10. What percentage of students did not prefer comedies?

62%

Algebra 1

Unit 12

Lesson 5 Homework

Name Answer Key

Date _____

Period _____

A study shows that the number of times a student skips breakfast has a negative correlation with student test scores.

1. What does it mean that there is a negative correlation?

Students with higher numbers of skipped breakfasts have lower test scores.

2. Does this mean that not eating breakfast causes students to do poorly on tests?

No

3. What other factors might contribute to this association?

-Running late/rushing and not properly focused

-Students may not be hungry in the morning and also struggle with test material

4. What other information would be helpful to determine whether not eating breakfast causes lower test scores?

-Are the tests only offered in the morning?

At a theme park, there is a positive correlation between the amount of rainfall and the number of ponchos sold.

5. What does it mean that there is a positive correlation?

The days with higher rainfall have more ponchos sold.

6. Does this mean that increased poncho sales leads to more rainfall? If not, what does it mean?

No. Some may have purchased ponchos because rain was forecasted.

7. What other factors might contribute to this association?

Some may have bought ponchos to protect them on the water rides.

At a ski resort, as snow accumulation increases, so does the injury rate.

8. What kind of correlation does this association represent?

Positive correlation

9. Is there evidence of a causal relationship?

yes

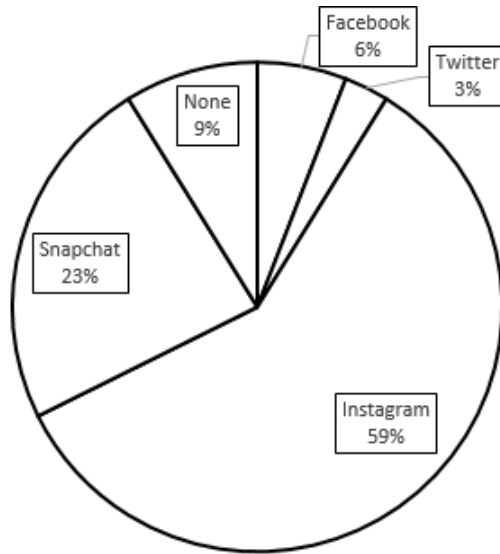
10. What other information would be helpful to evaluate the claim of a causal relationship?

What are the injuries and how are they specifically related to snowfall?

Algebra 1
Unit 12
Lesson 6 Homework

Name Answer Key
Date _____
Period _____

A circle graph is shown, with the results of a survey of 9th grade students about which social media platforms they use, with a margin of error of 1.2%. There are 475 9th graders in a particular school.



- Based on this survey, what is the estimated number of 9th graders who have an Instagram account?
 $0.59 \cdot 475 \approx 280$ ninth graders
- Based on this survey, what is the minimum number of 9th graders that would be expected to have a Facebook account?
 $6\% - 1.2\% = 4.8\%$ $0.048 \cdot 475 \approx 23$ ninth graders
- Based on this survey, what is the maximum number of 9th graders that would be expected to have a Twitter account?
 $3\% + 1.2\% = 4.2\%$ $0.042 \cdot 475 \approx 20$ ninth graders
- Based on this survey, what is the range of 9th graders that would be expected to have a Snapchat account?
 $23\% + 1.2\% = 24.2\%$ $0.242 \cdot 475 \approx 115$
 $23\% - 1.2\% = 21.8\%$ $0.218 \cdot 475 \approx 104$ Approximately 104 to 115.
- Based on this survey, what is the range of 9th graders that would be expected to have no social media accounts?
 $9\% + 1.2\% = 10.2\%$ $0.102 \cdot 475 \approx 48$ Approximately 37 to 48
 $9\% - 1.2\% = 7.8\%$ $0.078 \cdot 475 \approx 37$

A survey of parents indicated that 62% of parents felt their students spent too much time on social media, with a margin of error of 2%.

6. Out of a group of 2400 parents, how many parents would likely feel that their children spend too much time on social media?

$$0.62 \cdot 2400 = 1,488 \text{ parents} \quad 0.60 \cdot 2400 = 1440 \quad 0.64 \cdot 2400 = 1536$$

1440 to 1536 parents

7. Out of a group of 2400 parents, how many parents would likely **not** feel that their children spend too much time on social media?

$$0.38 \cdot 2400 = 912 \text{ parents} \quad 0.36 \cdot 2400 = 864 \quad 0.40 \cdot 2400 = 960$$

864 to 960 parents

In a survey of 200 families, 32% of families reported that they eat out less than twice a week, 45% report they eat out twice a week, and 23% report they eat out more than twice a week. The survey has a margin of error of 0.8%.

8. What is the expected number of families that eat out twice a week?

$$0.45 \cdot 200 = 90 \text{ families}$$

9. What is the range of families that eat out less than twice a week?

$$32 + 0.8 = 32.8\% \quad 0.328 \cdot 200 \approx 66 \quad \text{Approximately 62 to 66 families}$$
$$32 - 0.8 = 31.2\% \quad 0.312 \cdot 200 \approx 62$$

10. What is the range of families that eat out more than twice a week?

$$23 + 0.8 = 23.8\% \quad 0.238 \cdot 200 \approx 47 \quad \text{Approximately 44 to 47 families}$$
$$23 - 0.8 = 22.2\% \quad 0.222 \cdot 200 \approx 44$$